MILITARY SPECIFICATION

STARTER, AIR TURBINE, ACCESSORY DRIVE COMBINATION, AIRCRAFT ENGINE, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for an aircraft turbine engine air turbine starter and accessory drives (ATS/AD).

1.2 Classification. ATS/ADs shall be of the model designations shown on the specification sheets.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-365	Silver Plating (Electrodeposited); General Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ENGINEERING SPECIFICATIONS AND STANDARDS DEPARTMENT (CODE 93) NAVAL AIR ENGINEERING CENTER, LAKEHURST, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2995

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SPECIFICATIONS (Continued)

Federal (Continued)

QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for
PPP-B-585	Box; Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-T-60	Tape, Packaging, Waterproof

Military

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
DOD-D-1000	Drawings, Engineering and Associated Lists
MIL-H-5440	Hydraulic Systems, Aircraft, Types I and II, Design and Installation Requirements for
MIL-C-5501	Cap and Plug, Protective, Dust and Moisture Seal, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-8879	Screw Threads, Controlled Radius Root With Increased Minor Diameter; General Specification for
MIL-T-10727	Tin Plating, Electrodeposited or Hot-dipped, for Ferrous and Nonferrous Metals
MIL-C-11796	Corrosion Preventive Compound, Petrolatum Hot Application

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SPECIFICATIONS (Continued)

Military (Continued)

MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold-application
MIL-L-23699	Lubricating Oil, Aircraft Turbine Engines, Synthetic Base
MIL-H-25475	Hydraulic System, Missile Design, Installation Tests and Date Requirements, General Requirements for
MIL-V-81995	Valve, Starter Control, Aircraft Engine, General Specification for

STANDARDS

Military

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-280	Definitions of Item Levels, Item Exchangeability, Models, and Related Terms
DOD-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-1523	Age Controls of Age-sensitive Elastomeric Material
MS3106	Connector, Plug, Electric, Straight, Solder Contacts, AN Type
MS17320	Quick Attach-Detach Mounting Devices for Aircraft Electric Equipment, Circular, 5-, 8-, and 10- inch

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STANDARDS (Continued)

Military (Continued)

MS33540	Safety Wiring and Cotter Pinning, General Practices for
MS33649	Bosses, Fluid Connection, Internal Straight Thread
MS33658	Fitting End, Hose Connection, Standard Dimensions for

NAVAL AIR SYSTEMS COMMAND

MIL BULL 544	List of Specifications and Standards (Book Fore)

AIR FORCE - NAVY AERONAUTICAL

AND 20002	Drive - Type XII Engine Accessory
AND 10262	Flange, Type XII, Accessory Mounting

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS STANDARDS

AS 468	Drive, Accessory, Five Inch Bolt Circle
AS 471	Flange, Accessory, Five Inch Bolt Circle
E527	Numbering, Metals and Alloys, UNS

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Specification sheets. The individual ATS/AD requirements shall be as specified herein and in accordance with the applicable Specification Sheets. In the event of conflict between the requirements of this specification and the Specification Sheet, the requirements of the Specification Sheet shall govern.

3.2 Qualification. ATS/ADs furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.3).

3.2.1 Data. Unless specified in the contract, data cited in this specification is only required as part of the qualification submittal.

3.2.2 Materials and parts. Materials and parts used in the manufacture of the ATS/ADs shall be of high quality, suitable for the purpose and shall conform to applicable Government specifications. Materials, parts, and processes shall conform to applicable Government specifications listed in MIL-BULL-544. Contractor or industry specifications may be used for materials or processes not covered by the documents listed in MIL-BULL-544, provided the specifications are approved by the qualifying activity and contain provisions for adequate tests.

3.2.3 Selection of specifications, standards and drawings. Specifications, standards and drawings applicable to the design and construction of the ATS/ADs shall be in accordance with those documents listed in MIL-BULL-544 approved by the Naval Air Systems Command and takes precedence in accordance with MIL-STD-143.

3.2.4 Packings. All packings shall be in accordance with the requirements of MIL-H-5440 or MIL-H-25475 as applicable. Age of synthetic rubber parts shall be controlled in accordance with MIL-STD-1523.

3.2.5 Metals. All metals used in the construction of the ATS/ADs except metals which are in constant contact with oil, shall be of a corrosion-resistant type or shall be suitably protected in accordance with 3.2.5.1. The use of dissimilar metals, as defined by MIL-sTD-889, shall be avoided wherever practicable.

3.2.5.1 Corrosion protection. Metals which do not inherently possess adequate corrosion-resistant characteristics shall be suitably protected, in accordance with the following subparagraphs, to resist corrosion which may result from such conditions as dissimilar metal combinations, moisture, salt spray, and high temperature deterioration as applicable. Treatments that tend to promote stress corrosion in surfaces that contain residual tensile stresses shall be avoided.

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3.2.5.2 Ferrous and copper alloys. Ferrous alloys requiring corrosion-preventive treatment, and all copper alloys except for parts having bearing surfaces, shall have a suitable electrodeposited metallic coating selected from the following list, except that tin, cadmium, and zinc plating shall not be used for internal parts or on internal surfaces in contact with oil or exposed to its vapors, nor where subject to abrasion. Where not indicated, class or type is at the option of the manufacturer.

Cadmium Plating	QQ-P-416-1, Type II, Class B
Zinc Plating	QQ-Z-325, Class II, Type II
Chromium	QQ-C-320
Nickel	QQ-N-290
Silver	QQ-S-365
Tin	MIL-T-10727, Type I

3.2.5.3 Aluminum alloys. Unless otherwise specified, all aluminum alloys shall be anodized in accordance with MIL-A-8625, except that in the absence of abrasive conditions, they may be coated with chemical film in accordance with MIL-C-5541. The exceptions noted will be subject to the approval of the qualifying activity.

3.2.5.4 Magnesium alloy. Magnesium or its alloy shall not be used.

3.3 Design.

3.3.1 General. The ATS/AD is intended to serve two functions when mounted on the engine. When the required compressed air is applied to the air inlet, the ATS/AD shall furnish sufficient torque to start the engine. After the engine is started, the ATS/AD shall provide accessory power through an accessory pad driven with a suitable overrunning device to prevent operation of those parts necessary for starting only. The ATS/AD shall incorporate a reverse overload release mechanism to protect the unit against an engaging mechanism failure during engine operation. The ATS/AD shall be of a compact design and of such construction as to withstand the normal severe usage encountered in service under the environmental conditions specified herein.

3.3.2 Environmental conditions. The unit shall be designed for satisfactory operation, without Impairment of function or change in adjustment, throughout the ambient temperature range specified on the Specification Sheet and shall not be damaged when exposed to temperatures specified thereon. Consideration shall be given in the design of the ATS/AD to satisfactory operation during and after exposure to any combination of the following conditions in worldwide operation: humidity, fungus, rain, snow, sleet, hail, ice-fog, mildew, salt-spray, ice, sand, and dust .

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3.3.2.1 Vibration and acceleration forces. The ATS/AD shall be capable of withstanding the acceleration forces and completing the vibration tests specified on the Specification Sheet.

3.3.2.2 Control valve operation. The ATS/AD shall be capable of operating with a remote air control valve having an opening and a closing rate as specified on the Specification Sheet. It shall also be capable of operating with the MIL-V-81995 control valve specified on the Specification Sheet.

3.3.3 Component parts. The ATS/AD shall consist of a turbine rotor, reduction gearing, an automatically operated engaging and disengaging mechanism for coupling with the engine drive, and a torque overload protective device. Automatic shutoff controls shall be provided.

3.3.4 Oil seal. The ATS/AD shall be designed and constructed to prevent the entrance of engine oil into it and the leakage of lubricants from it.

3.3.5 Lubrication. The ATS/AD shall be self-lubricated, including Internal splines, by one of the following lubricant(s):

a. MIL-L-23699 Oil.

b. MIL-approved grease(s).

c. MIL-approved grease(s) and MIL-L-23699 oil.

If MIL-L-23699 oil is used, the ATS/AD shall also be compatible with MIL-L-7808 Oil. Use of MIL-L-23699 oil may be limited to -40 F. MIL-L-7808 oil shall be suitable for use throughout the temperature range specified on the Specification Sheet. The ATS/AD shall be capable of completing the endurance test specified on the Specification Sheet without changing or adding any lubricant.

3.3.6 oil system. An oil reservoir shall be provided with a dipstick marked “FULL” and “ADD” to determine the oil level. The dipstick shall be of a self-locking type and shall be retained with the ATS/AD by means of a lanyard when removed from the reservoir. Two oil reservoir ports, one on each side of the ATS/AD, shall be provided for dipstick installation; the unused port shall be capped. Oil fill ports shall be provided. Either the dipstick ports or separate ports similarly located will be acceptable. One oil drain port shall be provided. Optimum dipstick and port location with respect to starter servicing shall be determined during the aircraft mockup of 4.5.19. No liquid lubricant shall issue from any lubricating system vent(s) during any mode of operation.

3.3.7 Part numbering of interchangeable parts. All parts having the same design activity FSCM (Federal Supply Code for Manufacturers) and part number shall be interchangeable as defined in MIL-STD-280. The item identification and part number requirements of DOD-STD-100 shall govern the design activity part numbers and changes thereto.

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3.3.8 Design and construction changes.

3.3.8.1 Material substitutions. Material substitutions shall be subject to the approval of the qualifying activity. The SAE Index of Aeronautical Material Specification and ASTM Standard E527 (SAE Designation J1086) Numbering Metals and Alloys (UNS) may be used as guides for material substitutions.

3.3.8.2 Changes in design. No changes shall be made in the design or materials of an ATS/AD model that has been qualified, in accordance with this specification, except when the changes are approved by the qualifying activity in accordance with DOD-STD-480.

3.3.8.3 Parts list. The parts list of the ATS/AD which successfully completes the required qualification tests shall constitute a requirement for production ATS/ADs. Departure from the approved ATS/AD parts list in features of design and construction shall be governed by the provisions herein for changes in design.

3.3.8.4 Responsibility for changes. Approval by the qualifying activity of changes from the qualification test ATS/AD does not relieve the contractor of full responsibility for any adverse results of such changes.

3.3.9 Threaded parts.

3.3.9.1 Straight screw threads. All conventional straight screw threads shall conform to the requirements of MIL-S-8879 except for electrical connections, thermocouple harness, fluid fittings and ground and cut threads incorporated in proprietary parts which do not have equivalent standards. All exceptions shall conform to MIL-S-7742.

3.3.9.2 Tapered pipe threads. Tapered pipe threads shall be in accordance with MIL-P-7105 and shall be only for permanently plugging drilled or cored openings.

3.3.9.3 Threaded inserts. All threaded connections In non-ferrous materials shall have steel inserts which are suitably protected from electrolytic corrosion. Fill and drain boss inserts shall be designed to permit the use of standard gaskets or seals and standard straight thread plugs.

3.3.9.4 Safetying. Provisions shall be made to prevent all screws and screw parts from accidental loosening. Such parts shall be safety wired or fitted with safety stop nuts, where applicable. All other connections liable to be loosened by vibration shall be safetied in a similar manner. If safetying is by wiring; it shall be in accordance with MS33540. The use of lock nuts which incorporate non-metallic inserts as a locking feature shall be avoided.

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3.4 Reliability and maintainability.

3.4.1 Reliability. The ATS/AD shall be designed and developed to achieve the highest operational reliability commensurate with the design requirements. Factors to be considered during the design and manufacture of the ATS/AD to assure in-service reliability are:

a. Simplicity of design.

b. Selection and application of reliable components.

c. Consideration of operational and environmental parameters.

d. Mechanical structures - adequate design stress margin on all parts.

e. Use of proven manufacturing techniques.

f. Rigid quality control procedures imposed throughout procurement, manufacturing, assembly, and testing to assure that maximum design reliability is built into the equipment.

3.4.2 Maintainability. The ATS/AD shall have a minimum number of parts. Its design shall permit easy assembly, disassembly, location of trouble sources, routine service checking, adjustment and maintenance by personnel having a minimum of training. The modular concept of assembly shall be utilized wherever possible in order to increase maintainability and reduce overhaul and maintenance costs. The design goal shall be such that any subassembly of the ATS/AD can be replaced without disassembly of the complete ATS/AD or removal of other components. Maintenance between overhaul periods shall be held to an absolute minimum.

3.5 Functional requirements. The ATS/AD shall be designed and constructed to satisfy the functional requirements specified herein when operating under rated conditions and under other conditions specified on the Specification Sheet. Following qualification tests the ATS/AD shall satisfactorily meet the teardown inspection (see 4.5.21).

3.5.1 Rated conditions and cutout speed. Rated conditions for the ATS/AD and cutout speed for the starter section of the ATS/AD shall be as specified on the Specification Sheet.

3.5.2 Air turbine rotor. The rotor of each ATS/AD shall be designed for operation on compressed air having the characteristics specified on the Specification Sheet. No rotor blade resonance shall be present at any speed throughout the operating range of the rotor.

3.5.3 Engagement. The ATS/AD shall be engaged on the accessory drive by means of an overrunning clutch. This engagement shall be automatic upon initiation of airflow and shall be accomplished with a minimum of shock. The ATS/AD drive shall not be dependent on the engine accessory drive for alignment.

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3.5.4 Disengagement and reengagement. Disengagement of the ATS/AD shall be rapid and complete at the end of a starting cycle. Reengagement of the ATS/AD drive with the engine drive shall be accomplished rapidly and without damage to the ATS/AD or engine even though the ATS/AD or the engine drives are rotating.

3.5.5 Rotation. The rotation of the ATS/AD pad “A” and pad “B” drives shall be as specified on the Specification Sheet.

3.5.6 Shear section. The ATS/ADs shall incorporate a shear section to prevent damage to both the ATS/AD (except for shear section failure) and the engine in the event an excessive ATS/AD drive torque is developed. The output drive shall be designed so that if it fails, the resulting parts will be retained within the ATS/AD and engine accessory drive. The ATS/AD output drive shear section shall incorporate provisions to prevent axial movement of the output drive. The shear section shall be designed to permit replacement of the shear section without disassembly of the ATS/AD. The maximum permissible static shear torque to be transmitted to the engine by the ATS/AD at pad “A” under all operating conditions, including engagement and reengagements, shall not exceed the value specified on the Specification Sheet.

3*5.7 Automatic shutoff control. The ATS/AD shall be equipped with a cutout switch that will open a normally closed electrical circuit when the ATS/AD output drive attains the speed specified on the Specification Sheet. The switch shall be designed to withstand the electrical load imposed by the control valve specified thereon, but in no case shall it be designed to withstand less than a surge current of 2.5 amperes at 28 volts inductive load at -65°F. The switch shall not be polarity sensitive.

3.5.8 Altitude. The ATS/AD shall be capable of meeting the altitude requirements specified on the Specification Sheet.

3.5.9 Attitude. The ATS/AD shall be capable of meeting the attitude requirements specified on the Specification Sheet.

3.5.10 Overrunning. The ATS/AD shall not be damaged as a result of overrunning for the time and at the speeds specified on the Specification Sheet. This overrunning shall be accomplished at the maximum misalignment specified for the engine accessory drive and without lubrication from the engine. The drag in the overrunning clutch shall be minimized to prevent rotation of components related to operation as a starter.

3.6 Performance. The ATS/AD performance shall meet the requirements specified on the Specification Sheet. All bidders and contractors shall furnish performance data on each model ATS/AD they propose to furnish to the Navy in accordance with the requirements of paragraph 4.4.1 herein. The data shall be in ‘the form shown on Figures 1 to 5, inclusive, and may be estimated or calculated until the ATS/AD has been tested. After that time, performance data shall be based on test results.

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3.7 Weight. The total weight of the ATS/AD, including lubricant, shall not exceed the value specified on the Specification Sheet.

3.8 Installation.

3.8.1 Size and dimensions. The size of the ATS/AD shall be within the envelope configuration as specified on the Specification Sheet. The ATS/AD installation configuration shall be as shown thereon and shall be subject to approval by the Navy. Manufacturers interested in qualifying an ATS/AD to this specification are urged to coordinate the installation with the Navy prior to fabrication of a qualification ATS/AD. The manufacturer's Navy-approved installation drawing will form a part of this specification.

3.8.2 Mounting. Mounting of the ATS/AD shall not require special tools for installation, nor shall there be any loose or separable parts when the unit is removed from the engine. The ATS/AD shall be designed to mount on the specified engine accessory drive within the specified tolerances, without requiring any measuring or adjustment to the accessory drive or to the ATS/AD prior to installation.

3.8.2.1 Engine accessory drive. The ATS/AD shall provide two pads, pad "A" and pad "B." The ATS/AD shall be mounted on the aircraft engine at pad "A" and shall be suitable for mating with the engine accessory drive specified on the Specification Sheet. Pads "A" and "B" shall be in compliance with the Specification Sheet.

3.8.2.2 Trial installation and engine cranking. The ATS/AD shall be compatible with regard to form, fit and function when installed on the applicable aircraft and aircraft engine (see 4.5.19 and 4.5.20).

3.8.3 Air inlet and exhaust connections. The ATS/AD air inlet connections and the ATS/AD exhaust shall be as described on the Specification Sheet.

3.8.3.1 Mechanical loads and moments. The ATS/AD air inlet and exhaust flanges shall be capable of withstanding the following loads and moments resulting from thermal expansion, deflection, and vibration when installed in aircraft:

a. Shear - 50 lbs.

b. Axial - 100 lbs.

c. Moment - 300 lbs-in.

3.8.3.2 Air inlet pressure loads. The ATS/AD air inlet flange shall be capable of withstanding the tensile load resulting from the maximum air pressure specified on the Specification Sheet.

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3.8.3.3 Extraneous loads and moments. The ATS/AD air inlet flange shall be capable of withstanding, simultaneously, the acceleration forces defined in paragraph 3.3.2.1, the tensile load defined in paragraph 3.8.3.2, and the mechanical loads and moments defined in paragraph 3.8.3.1.

3.9 Marking.

3.9.1 Nameplate. Each ATS/AD shall be clearly and permanently marked with an irreversible nameplate in accordance with MIL-STD-130, securely attached to a part of the ATS/AD which will not ordinarily be renewed during normal service life, specifying the following information:

STARTER, AIR TURBINE, ACCESSORY DRIVE
COMBINATION, AIRCRAFT ENGINE, GENERAL
SPECIFICATION FOR
Specification: MIL-S-22518(AS)
Part Number: M22518/*
**Rotation: (Indicate by an arrow)
Fill with F1. oz., MIL-L-23699 or MIL-L-7808
Manufactures part number _____ *** LESS ISSUE _____
Manufacturer's serial number _____
Government order or contract number _____
Federal Stock Number _____

NOTE:
* Insert correct slash sheet number

** The arrow showing the direction or rotation may be included on a single nameplate with other specified data or, at the option of the contractor, it may be stamped on a separate plate attached immediately adjacent to the main nameplate.

*** For use by Naval Air Rework Facilities

3.9.2 Marking of product. The specification and M22518/*Part Number shall not be used on ATS/AD until notification has been received in writing that the ATS/AD has been approved for use in Naval aircraft.

3.9.3 Marking of parts. Each part and assembly shall be marked with its part number, and drawing revision symbol where applicable, which shall be the same as the drawing number, except for the following:

a. Those parts which do not have a suitable or sufficient surface for a part number.

b. Those parts which are permanently assembled by welding, brazing, soldering, or riveting shall carry the assembly part number.

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3.9.3.1 Part number. The part number shall, when practicable, be located to permit its being read after assembly of the complete unit. The marking shall be such that it will not be effaced or obliterated as a result of service usage during the life of the part.

3.10 Workmanship. The workmanship displayed in fabrication and assembly of the ATS/AD shall be such as to assure, within design limitations, the ability of the ATS/ADs to meet their performance requirements under all applicable environmental conditions specified herein. Unauthorized repair, welding, heavy burrs, or parts assembled by introduction of high stresses not prescribed in the drawings, are typical signs of inferior workmanship and shall be cause for rejection. The standards of workmanship exhibited in the approved ATS/AD sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship (see 4.4.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the ATS/AD shall be classified as follows:

a. Qualification tests (4.3).

b. Quality conformance inspection (4.4),

4.2.1 Retention of qualification. Retention of qualification consists of a periodic verification to determine compliance of the qualified product with the requirements of this specification and applicable Specification Sheet (see 3.2 and 4.3).

4.3 Qualification tests.

4.3.1 Qualification test samples. The qualification test sample shall consist of two ATS/ADs and one additional output shaft coupling per ATS/AD of each manufacturer's part number upon which qualification is desired. These ATS/ADs and one additional output shaft coupling per ATS/AD shall be accompanied by one complete parts list, one complete set of detail and assembly drawings and a qualification test report. The drawings shall be in accordance with DOD-D-1000. Each qualification

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test sample shall be plainly identified by securely attached tags clearly and durably marked with the following information, and shall be forwarded to the Commanding Officer, Naval Air Propulsion Center, Attn: PE62, Trenton, New Jersey 08628, after obtaining authorization from the qualifying activity for submittal of the qualification test samples:

Sample for Qualification Test
Submittal by (name) (date) for qualification test in accordance with the requirements of Specification MIL-S-22518A(AS) (date) under authorization (reference letter authorizing tests).

Starter, Air Turbine, Accessory Drive, Combination, Aircraft Engine, General Specification for
Manufacturer's Part Number
Name of Manufacturer
Specification Sheet Number or other identification
Navy Work Unit Assignment Number

4.3.1.1 Qualification test report. The supplier shall furnish performance data on each model ATS/AD they propose to have qualified to the qualifying activity in accordance with the requirements of 4.3.1 herein. The data shall be in the form shown on Figures 1 through 5, and may be estimated or calculated until the ATS/AD has been tested. After that time, performance data shall be based on test results. The report shall demonstrate that the size of the ATS/AD is within the envelope configuration specified on the applicable Specification Sheet. The ATS/AD installation configuration shall be as shown thereon and shall be subject to approval by the qualifying activity. Manufacturers interested in qualifying ATS/ADs to this specification are required to coordinate installation with the qualifying activity prior to fabrication of qualification units. The manufacturer's approved installation drawing shall form a part of this specification.

4.3.1.2 Drawings and diagrams. The manufacturer shall furnish the following drawings and diagrams to the qualifying activity with the submission of the qualification test samples.

a. ATS/AD installation drawing - the drawing shall show ATS/AD center of gravity, all connecting fittings, clearance required for installation and removal, filter locations and cleaning instructions if applicable.

b. Electrical diagram - the electrical diagram shall show all electrical circuits up to the connecting points of the aircraft electrical system.

c. A complete cutaway or cross-sectional drawing of the ATS/AD and all components showing all parts in their normal assembled position and shall specify part numbers of all parts or components subassemblies. The following data shall be furnished on or together with the assembly drawing:

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1. All dimensions, clearances, and fits.

2. Materials, treatment and finishes (identify each by specification number).

3. Construction.

4. Characteristics and ratings.

d. A complete set of level 3 production drawings and associated lists in accordance with DOD-D-1000.

e. Packing requirements and packing drawings for the ATS/AD.

f. ALL contractor specifications listed in the manufacturing drawings.

4.3.2 Tests. The qualification tests shall consist of the tests specified under 4.5 and listed in Table I, conducted on the applicable specimens. The tests to be conducted on each ATS/AD shall be determined by the qualifying activity. If considered desirable by the qualifying activity, the qualification test may be supplemented with tests under actual or simulated service conditions, provided they are not in contradiction to the requirements specified herein. The qualifying activity reserves the right to disassemble the ATS/ADs partially or completely before, during, and after the qualification test.

4.3.3 Retention. The retention of qualification shall consist of periodic verification to determine compliance of the qualified ATS/AD with the requirements of this specification. Verification shall be conducted at intervals not exceeding 2 years.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the following:

Individual inspection (4.4.1).

4.4.1 Individual inspection. The inspection test of the ATS/AD shall consist of the tests described herein for the following:

a. Rated torque calibration test (4.5.9).

b. Overrunning speed test (4.5.10).

c. The airflow at the stall condition shall be measured (4.5.4). During this check, rotation of the ATS/AD output shaft shall not exceed 200 rpm.

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TABLE I, Qualification tests.

TEST	PARAGRAPH
Examination of product	4.4.2
Initial calibration	4.5.4
No-load operation	4.5.5
No-load operation (acceptance)	4.5.6
Endurance test	4.5.7
Post-endurance calibration	4.5.8
Rated torque calibration	4.5.9
Overrunning	4.5.10
Extreme temperature operation	4.5.11
Consecutive cycle operation	4.5.12
Sustained motoring	4.5.13
Vibration	4.5.14
Disengagement and reengagement	4.5.15
Inverted flight	4.5.16
Altitude accessory drive	4.5.17
Shear section strength	4.5.18
Engine cranking	4.5.20
Teardown inspection	4.5.21
Rejection and retest	4.5.22

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d. The cutoff switch setting shall be checked (3.5.1).

e. Examination of product (4.4.2).

In addition to the above-listed tests, the ATS/ADs may be subjected to any other tests specified herein recommended by the Inspector and authorized by the Navy to determine conformance with the requirements of this specification.

4.4.2 Examination of product. Each ATS/AD shall be carefully examined to determine conformance with this specification with respect to all requirements not covered by specific tests.

4.5 Tests.

4.5.1 Test conditions. Unless otherwise specified, all tests shall be conducted with the ATS/AD at an ambient temperature range of 50 F to 120°F at rated air inlet conditions as specified on the Specification Sheet and with the exhaust conditions at the prevailing laboratory atmospheric pressure. Tests may be conducted in the order deemed most desirable by the testing agency.

4.5.1.1 Pressure rate increase. Unless otherwise specified, the initial instantaneous pressure rise shall be limited to the maximum opening rate specified on the Specification Sheet, and thereafter the pressure rise shall be smooth and at a positive rate. Zero time shall correspond to the first indication of pressure rise measured three inches upstream of the ATS/AD inlet flange.

4.5.2 Test apparatus.

4.5.2.1 Speed. ATS/AD output drive speeds shall be measured by an apparatus with an accuracy of ±1 percent.

4.5.2.2 Airflow. Airflow measurements shall be made using flowmeter conforming to ASME report: Fluid Meters, Their Theory and Applications, Sixth Edition.

4.5.2.3 Torque. Torque measurements shall be made by means of a torquemeter with an accuracy of ±1 percent.

4.5.2.4 Temperature. All temperature measurements shall be reported in degrees Fahrenheit with an accuracy of ±5°F.

4.5.2.5 Pressure. All air and gas pressures shall be reported in inches of mercury absolute or pounds per square inch absolute, whichever is applicable, and shall be accurate within ±1 percent.

4.5.3 Instrumentation. Instrumentation shall be provided during qualification and inspection tests to observe or determine the following ATS/AD values:

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a. ATS/AD output shaft speeds.

b. ATS/AD output torque.

c. Air turbine airflow.

d, Air turbine inlet total pressure.

e. Air turbine inlet total temperature.

*f l Air turbine outlet total pressure.

*g. Air turbine outlet static pressure.

*h. Air turbine outlet total temperature.

*1. Oil sump temperature.

*NOTE; These parameters need not be measured during the inspection tests unless required to determine acceptable performance.

4.5.4 Initial calibration. Initial calibration shall consist of starter-type operation at stabilized pad "A" speeds ranging from zero to runaway speed in equal increments to provide a minimum of six speeds. The items listed in paragraph 4.5.3 shall be recorded. Stabilized speeds shall be obtained by loading pad "A" by means of a suitable loading device. The airflow versus speed curve shall be within the limits specified on the Specification Sheet. The pad "A" torque versus speed tune shall be within the limits specified on the Specification Sheet. External cooling may be used during this calibration if excessive oil temperatures will result from prolonged operation required to obtain stabilized data. Pad "B" shall not be loaded.

4.5.5 No-load operation (qualification). With the cutout switch made inoperative, the ATS/AD shall be operated as a starter at a stabilized no-load speed for a period of time specified on the Specification Sheet and then shut down. The no-load speed obtained during this test shall be not less than that specified on the Specification Sheet. There shall be no evidence of damage or failure upon completion of the test.

4.5.5.1 Burst test. The ATS/AD shall be burst tested at no-load with an inlet air total pressure and at an inlet air total temperature specified on the Specification Sheet. Sufficient airflow shall be supplied at the above air inlet conditions until the ATS/AD has reached stabilized free-running speed at which time the airflow shall be shut off* The turbine wheel shall not fail during this test. Pad "B" shall not have an accessory installed thereon during this test.

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4.5.6 No-load operation (acceptance). With the cutout switch made inoperative, the ATS/AD shall be brought to a stabilized speed operating as a starter and then shut down. The no-load speed obtained during this test shall be not less than that specified on the Specification Sheet. There shall be no evidence of damage or failure upon completion of the test.

4.5.7 Endurance test. The ATS/AD shall complete the endurance test specified on the Specification Sheet without damage to, adjustment of, or replacement of any of the component parts. The cutout switch shall be loaded electrically by the type of control valve specified on the Specification Sheet. The cutout switch shall function within the limits specified on the Specification Sheet.

4.5.8 Post-endurance calibration. After the completion of the endurance test specified on the Specification Sheet and at ATS/AD inlet air conditions and speeds corresponding to those obtained during the initial calibration, the airflow shall remain within the maximum specified limit and the torque shall not have decreased more than 10 percent as a result of the endurance testing.

4.5.9 Rated torque calibration (acceptance). The ATS/AD shall be operated as a starter at rated drive torque. The minimum output drive speed and the maximum airflow shall be within the limits specified on the Specification Sheet.

4.5.10 Overrunning. This test shall consist of driving the pad "A" drive in accordance with the conditions noted on the Specification Sheet. No damage or excessive wear of the ATS/AD parts shall result from this test.

4.5.11 Extreme temperature operation.

4.5.11.1 Low temperature operation. The ATS/AD temperature shall be maintained at the value specified on the Specification Sheet and the ATS/AD operated as specified thereon. The ATS/AD shall not be damaged as a result of low temperature soaking or operation.

4.5.11.2 High temperature operation. The ATS/AD temperature shall be maintained at the value specified on the Specification Sheet and operated as specified thereon. The ATS/AD shall not be damaged as a result of high temperature soaking or operation.

4.5.12 Consecutive cycle operation. The ATS/AD shall be operated as a starter for the number of consecutive endurance test cycles specified on the Specification Sheet without damage. Time between deceleration to rest and initiation of the next cycle shall not exceed 30 seconds. External cooling shall not be permitted during this test.

4.5.13 Sustained motoring. While coupled to a dynamometer or other power-consuming device, the ATS/AD shall be operated as a starter at rated torque for the time specified on the Specification Sheet. No damage to the ATS/AD shall result from this test. External cooling shall not be permitted during this test. This test shall be conducted only when required by the applicable Specification Sheet.

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4.5.14 Vibration. The ATS/AD shall be subjected to the vibration test of MIL-STD-810 specified on the Specification Sheet. No damage to the ATS/AD shall result from this test.

4.5.15 Disengagement and reengagement test. Disengagement and reengagement tests shall be conducted on the ATS/AD functioning as a starter to demonstrate compliance with 3.5.4 herein. It shall be disengaged and reengaged at speeds ranging from zero to cutout speeds in equal Increments selected to provide a minimum of 10 engagements. No damage to the ATS/AD shall result from this test.

4.5.16 Inverted flight test. When full of oil and ready for service, the ATS/AD shall be mounted in an inverted position for the minimum of time specified on the Specification Sheet. Any oil leakage in excess of one drop at any location, particularly at the vents, shall be considered unacceptable.

4.5.17 Altitude accessory drive tests. The ATS/AD shall undergo an accessory load cycle test at conditions simulating a typical flight profile of an aircraft in which it will be installed. The altitude test shall be conducted in accordance with the test schedule shown on the Specification Sheet. The ATS/AD shall perform satisfactorily during the altitude tests without excessive oil consumption, oil leakage or foaming of the oil.

4.5.18 Shear section strength. After completion of all tests, both shear shafts of the two qualification ATS/ADs, submitted in accordance with paragraph 4.3.1 herein, shall be sheared and the shearing section torque recorded. The shear value shall not exceed the value specified on the Specification Sheet.

4.5.19 Mockup. The supplier shall conduct a mockup or trial installation to verify form, fit, and function of the product as required by this specification and to demonstrate compatibility with the applicable aircraft. Prior to submitting the ATS/ADs to the Government for qualification testing, the suppliers should arrange, with the Naval Air Systems Command (AIR-536), a mockup or trial installation of the product that is offered to the Navy for qualification testing. Any costs incident to the actual mockup or trial installation on the aircraft is to be borne by the offeror.

4.5.20 Engine cranking tests. At the option of the Navy, the ATS/AD must complete 75 actual starts and 5 hours of running on the engine for which it was designed. The ATS/AD may be one submitted for qualification or it may be an additional model furnished by the manufacturer for such tests. The tests may be performed in an engine test cell at the Naval Air Propulsion Center or at the engine manufacturer's plant, in a prototype aircraft, or in an aircraft specifically designed for such tests. When using an ATS/AD submitted for qualification test, the 75 starts and 5 hours running shall be included in the endurance test. Test conditions shall be at the prevailing ambient conditions where the tests are performed and the air source shall be of an approved Navy type.

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4.5.21 Teardown inspection. A teardown inspection consisting of complete disassembly of the ATS/AD shall be made at the completion of the qualification test. Dimensional inspection shall be made of all wearing parts. Rotating parts shall be subjected to Magnaflux or Zyglo inspection, except that assembled anti-friction bearings shall not be subjected to Magnaflux inspection. Rotating parts subjected to centrifugal stresses shall be measured on dimensions likely to show growth. Where practicable, seal clearances shall be determined. Failure of or any abnormal wear or damage to any part, sufficient to impair continued use of the ATS/AD shall be sufficient cause for rejection of the ATS/AD.

4.5.22 Rejection and retest. Any ATS/AD failing to meet the requirements of the qualification tests shall be rejected and returned at the contractor's expense. ATS/ADs which have been rejected may be replaced or repaired to correct the defects and resubmitted for all specified tests. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the original defects shall be furnished. ATS/ADs rejected after retest shall not be resubmitted without the specific approval of the qualifying activity.

5* PREPARATION FOR DELIVERY

5.1 Application. The requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation. All ports shall be sealed by closures conforming to Specification MIL-C-5501. All exterior surfaces of the component shall be protected by corrosion-preventative compound conforming to Specification MIL-C-11796 or MIL-C-16173. The component shall then be wrapped or bagged in grade A, greaseproof paper conforming to Specification MIL-B-121 and sealed with tape conforming to Specification PPP-T-60.

5.3 Packaging.

5.3.1 Level A. Unless otherwise specified, each ATS/AD shall be packaged for shipment in accordance with MIL-P-116. Method IIb of this specification shall be employed for all ATS/ADs designated as installation ATS/AD. Method IId of this specification shall be employed for all ATS/ADs designated as spare ATS/ADs.

5.3.2 Level C. ATS/ADs shall be packaged in accordance with standard commercial practice.

5.4 Packing. Shipping containers, insofar as practicable, shall contain an identical number of articles, shall be of a uniform size, and shall be designed to enclose the contents in a snug, tight-fitting manner. When packed for shipment, the gross weight of the exterior container shall not exceed 200 pounds. The above applies to 5.3.1 and 5.3.2.

5.4.1 Level A, ATS/ADs packaged in accordance with 5.3.1 shall be packed in an exterior shipping container in accordance with PPP-B-601 (overseas type) or PPP-B-621 (overseas type).

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5.4.2 Level B. ATS/ADs packaged as specified in 5.3.1 shall be packed in an exterior shipping container conforming to PPP-B-601 (domestic type) or PPP-B-585.

5.4.3 Level C. The interior packages shall be placed in substantial commercial exterior shipping containers constructed to insure acceptance by common or other carrier for safe transportation at the lowest rate, to the point of delivery. Except as specified herein, the container shall conform to the requirements of the Uniform Freight Classification Rules in effect at the time of shipment.

5.5 Marking. Each package and exterior shipping container shall be marked to comply with MIL-STD-129 and to include the following:

STARTER, AIR TURBINE, ACCESSORY DRIVE COMBINATION,
AIRCRAFT ENGINE, GENERAL SPECIFICATION FOR
Specification MIL-S-22518A(AS) '
Specification Sheet MIL-s-22518/*
Mfg. Model Designation
Ser. No.
Date of Manufacture
Contract No.
NSN

NOTE: *Insert applicable slash number.

6. NOTES

6.1 Intended use. These ATS/ADs are intended for use in starting aircraft gas turbine engines.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, date of this specification, and applicable Specification Sheet title and part number.

b. Selection of applicable levels of preservation, packaging and packing (5).

c. Power and speed limiting device, Type II starters (3.3.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification.

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The activity responsible for the Qualified Products List is the Naval Air Systems Command, Attn: AIR-53641, Department of the Navy, Washington, DC 20360; however, information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Air Propulsion Center, Attn: PE62, Trenton, NJ 08628.

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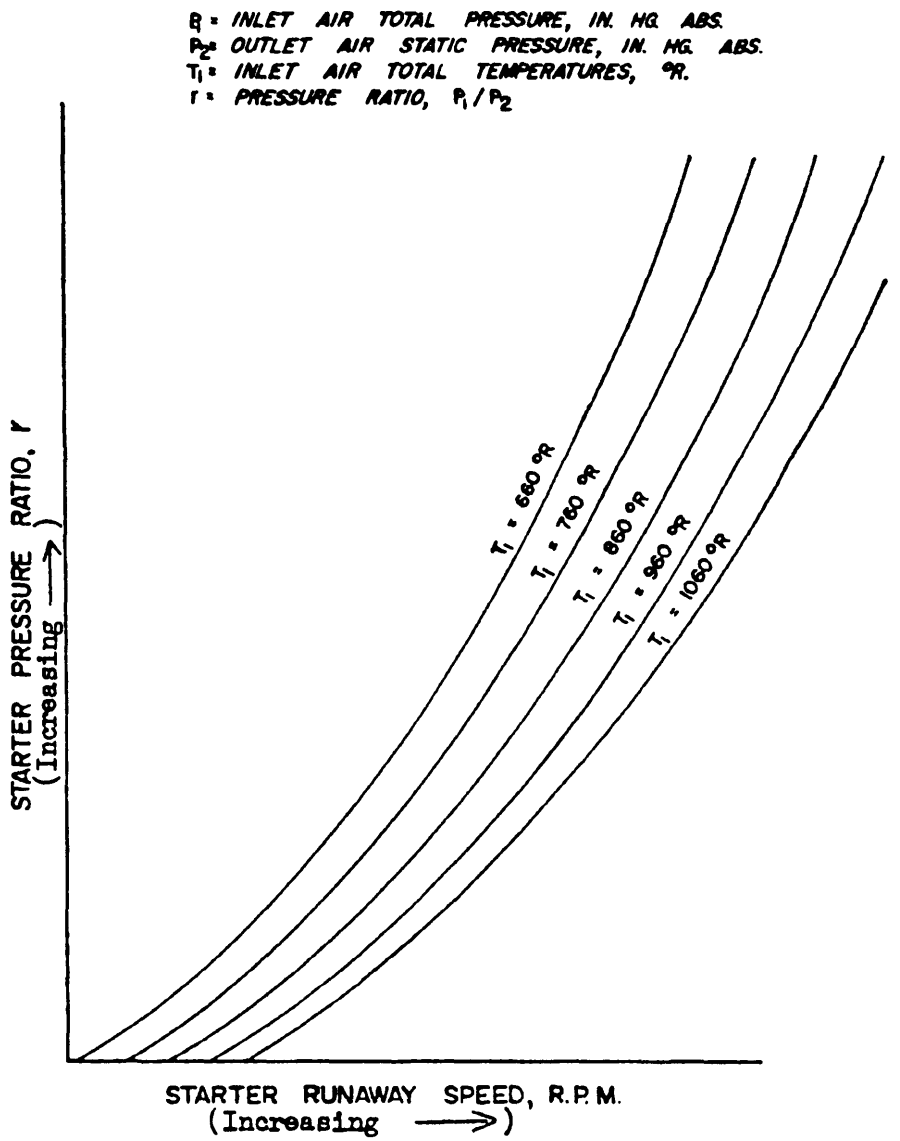


FIGURE 1. Starter no-load speed.

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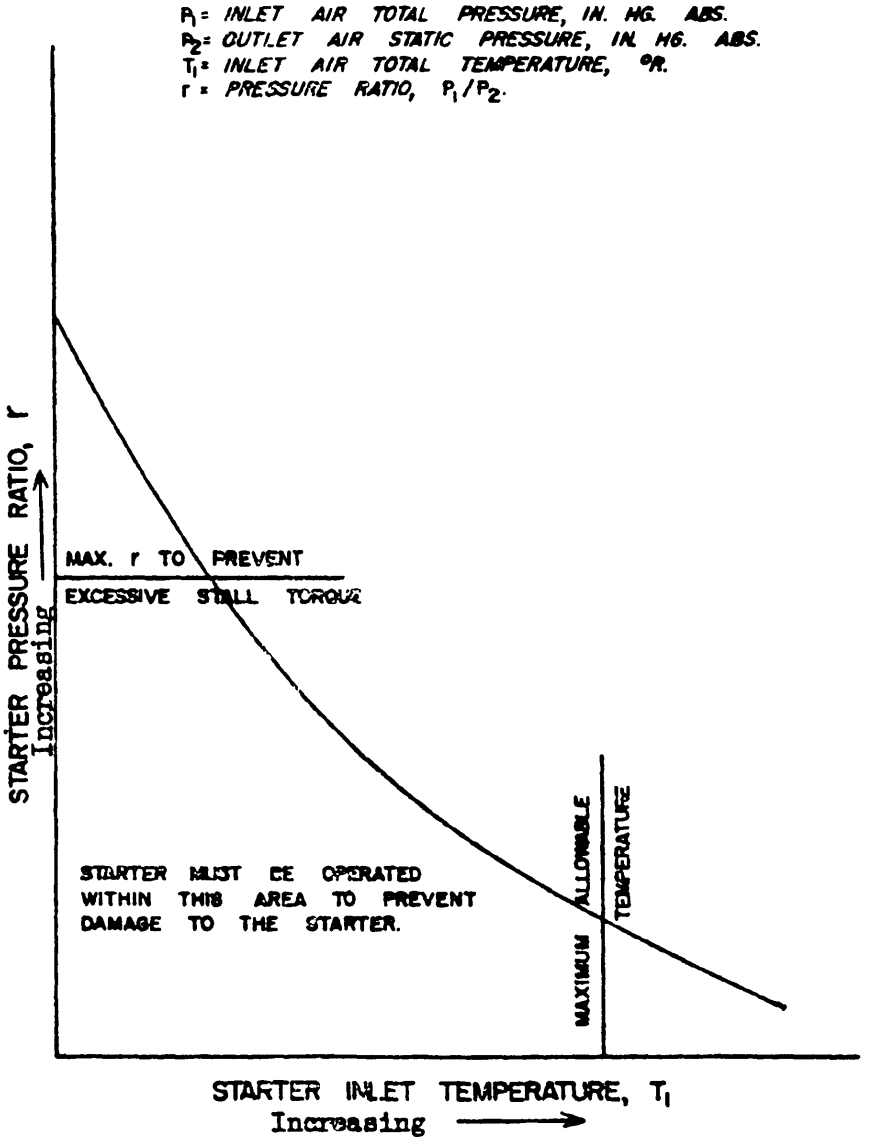


FIGURE 2. Pressure ratio limit to prevent starter overspeed.

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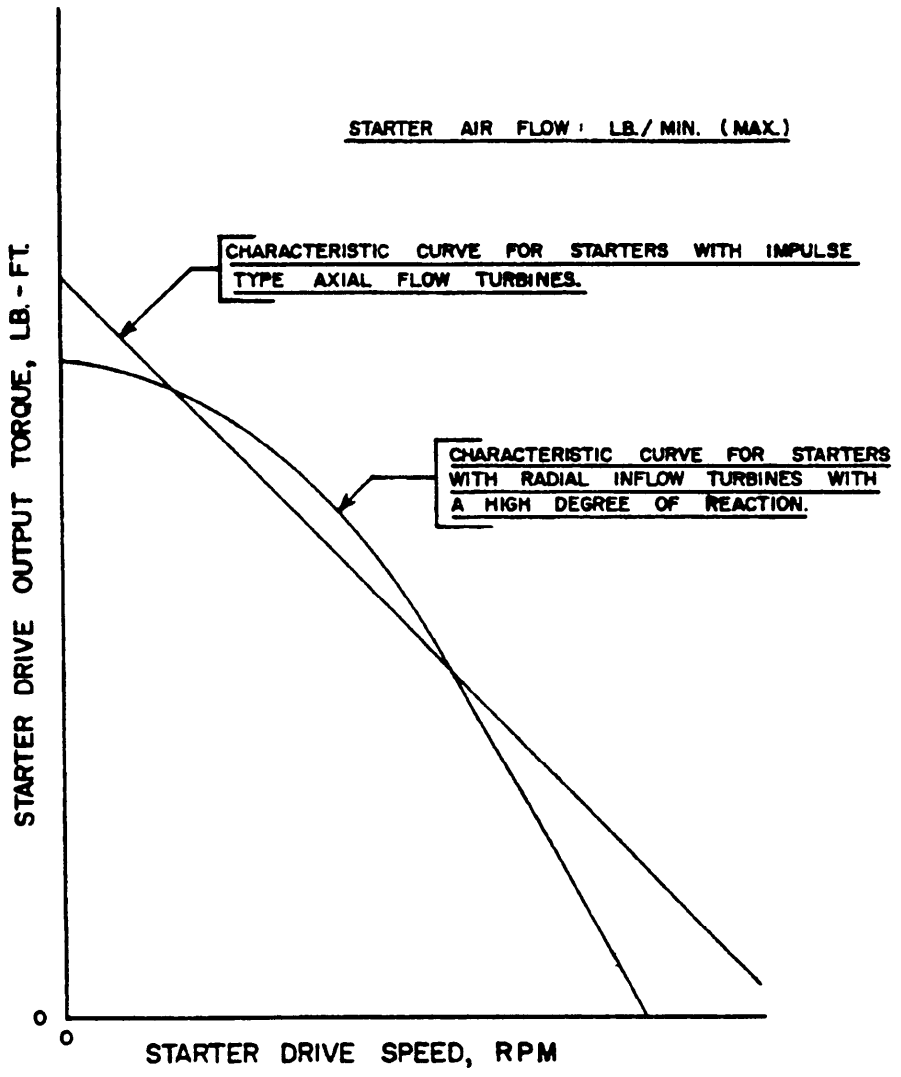


FIGURE 3. Starter performance under rated inlet conditions.

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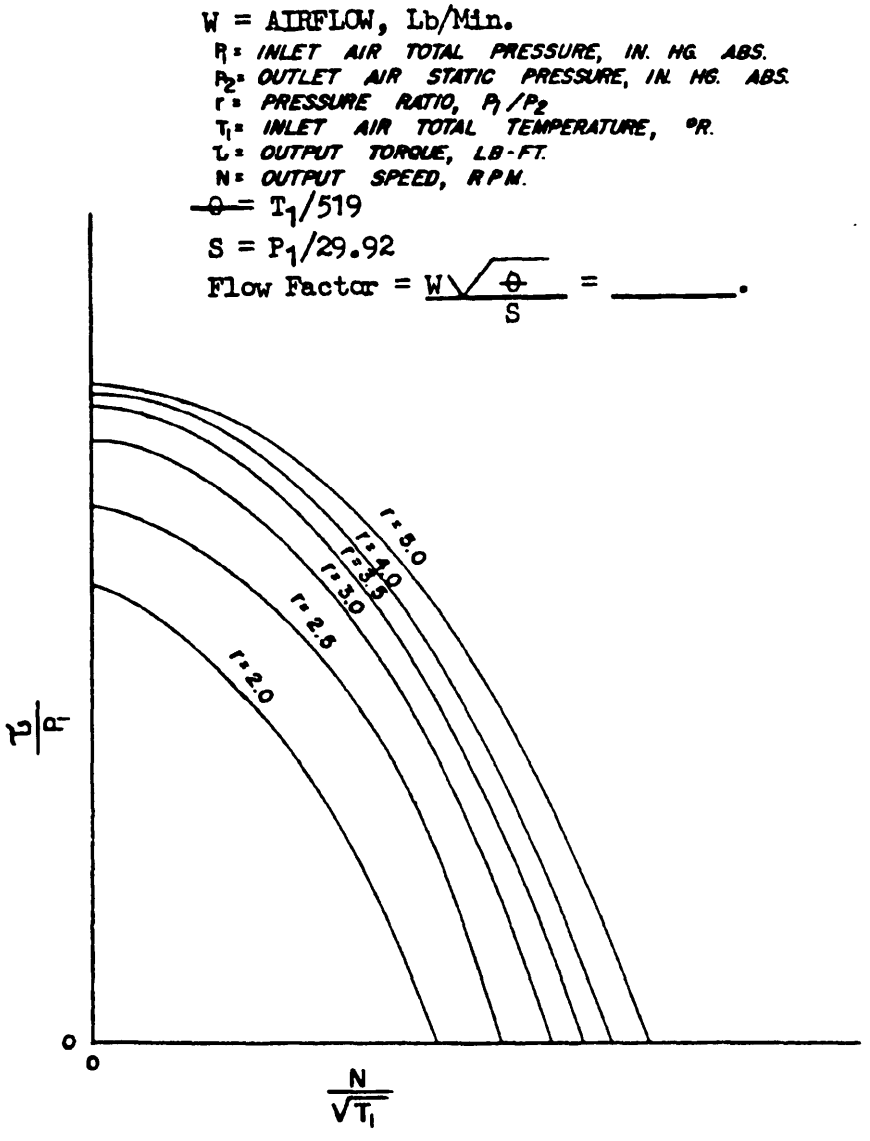


FIGURE 4. Starter performance under range of inlet conditions.

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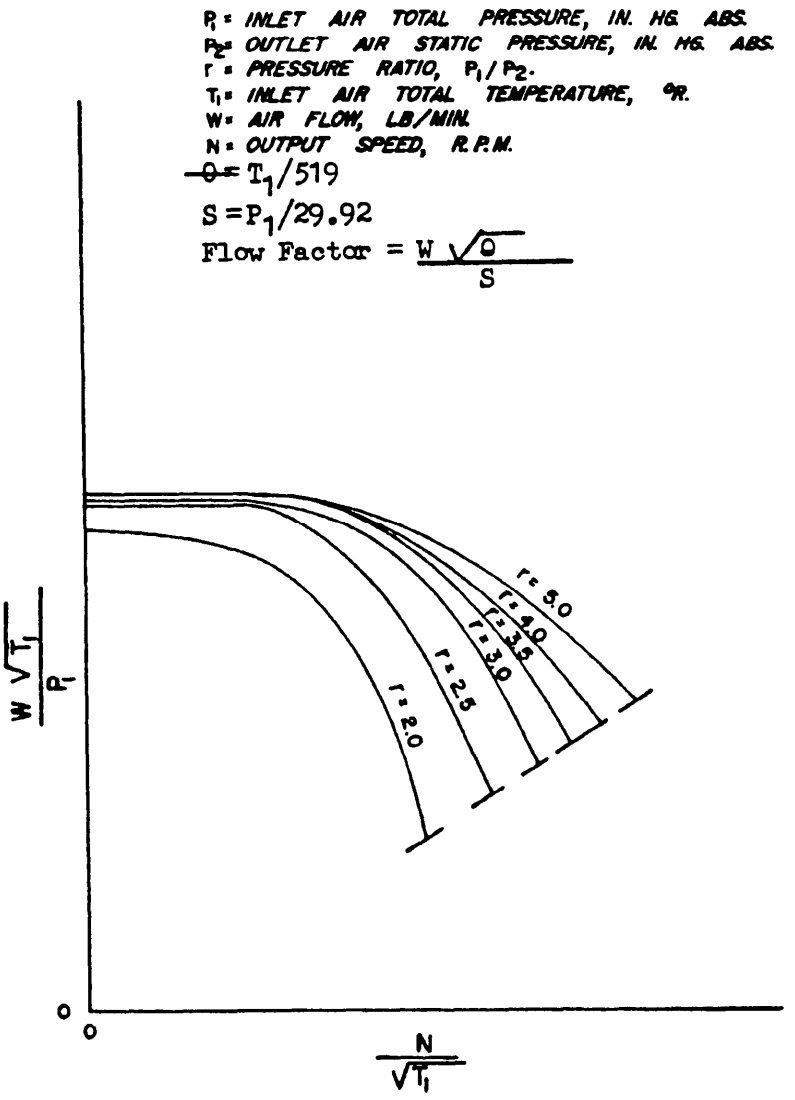


FIGURE 5. Starter performance under range of inlet conditions.

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NUMERICAL INDEX

Section		Page
1.	SCOPE	1
1.1	Scope	1
1.2	Classification	1
2.	APPLICABLE DOCUMENTS	1 - 4
3.	REQUIREMENTS	5 - 13
3.1	Specification sheets	5
3.2	Qualification	5
3.2.1	Data	5
3.2.2	Materials and parts	5
3.2.3	Selection of specifications, standards and drawings	5
3.2.4	Packings	5
3.2.5	Metals	5
3.2.5.1	Corrosion protection	5
3.2.5.2	Ferrous and copper alloys	6
3.2.5.3	Aluminum alloys	6
3.2.5.4	Magnesium alloys	6
3.3	Design	6
3.3.1	General	6
3.3.2	Environmental conditions	6
3.3.2.1	Vibration and acceleration forces	7
3.3.2.2	Control valve operation	7
3.3.3	Component parts	7
3.3.4	Oil seal	7
3.3.5	Lubrication	7
3.3.6	oil system	7
3.3.7	Part numbering of interchangeable parts	7
3.3.8	Design and construction changes	8
3.3.8.1	Material substitutions	8
3.3.8.2	Changes in design	8
3.3.8.3	Parts list	8
3.3.8.4	Responsibility for changes	8
3.3.9	Threaded parts	8
3.3.9.1	Straight screw threads	8
3.3.9.2	Tapered pipe threads	8
3.3.9.3	Threaded inserts	8
3.3.9.4	Safetying	8
3.4	Reliability and maintainability	9
3.4.1	Reliability	9
3.4.2	Maintainability	9
3.5	Functional requirements	9
3.5.1	Rated conditions and cutout speed	9
3.5.2	Air turbine rotor	9
3.5.3	Engagement	9

Source: https://assist.dla.mil - Downloaded: 2026-10-19T12:43Z
Check the source to verify that this is the current version before use.

NUMBERICAL INDEX (Continued)

Section		Page
3.5.4	Disengagement and Reengagement	10
3.5.5	Rotation	10
3.5.6	Shear section	10
3.5.7	Automatic shutoff control	10
3.5.8	Altitude	10
3.5.9	Attitude	10
3.5.10	Overrunning	10
3.6	Performance	10
3.7	Weight	11
3.8	Installation	11
3.8.1	Size and dimensions	11
3.8.2	Mounting	11
3.8.2.1	Engine accessory\ drive	11
3.8.2.2	Trial installation and engine cranking	11
3.8.3	Air inlet and exhaust connections	11
3.8.3.1	Mechanical loads and moments	11
3.8.3.2	Air inlet pressure loads	11
3.8.3.3	Extraneous loads and moments	12
3.9	Marking	12
3.9.1	Nameplate	12
3.9.2	Marking of product	12
3.9.3	Marking of parts	12
3.9.3.1	Part number	13
3.10	Workmanship	13
4.	QUALITY ASSURANCE PROVISIONS	13 - 21
4.1	Responsibility for inspection	13
4.2	Classification of tests	13
4.2.1	Retention of qualification	13
4.3	Qualification tests	13
4.3.1	Qualification test samples	13
4.3.1.1	Qualification test report	14
4.3.1.2	Drawings and diagrams	14
4.3.2	Tests	15
4.3.3	Retention	15
4.4	Quality conformance inspection	15
4.4.1	Individual inspection	15
4.4.2	Examination of product	17
4.5	Tests	17
4.5.1	Test conditions	17
4.5.1.1	Pressure rate increase	17
4.5.2	Test apparatus	17
4.5.2.1	Speed	17
4.5.2.2	Airflow	17
4.5.2.3	Torque	17
4.5.2.4	Temperature	17
4.5.2,5	Pressure	17

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Check the source to verify that this is the current version before use.

NUMERICAL INDEX (Continued)

Section		Page
4.5.3	Instrumentation	17
4.5.4	Initial calibration	18
4.5.5	No-load operation (qualification)	18
4.5.5.1	Burst test	18
4.5.6	No-load operation (acceptance)	19
4.5.7	Endurance test	19
4.5.8	Post-endurance calibration	19
4.5.9	Rated torque calibration (acceptance)	19
4.5.10	Overrunning	19
4.5.11	Extreme temperature operation	19
4.5.11.1	Low temperature operation	19
4.5.11.2	High temperature operation	19
4.5.12	Consecutive cycle operation	19
4.5.13	Sustained motoring	19
4.5.14	Vibration	20
4.5.15	Disengagement and reengagement test	20
4.5.16	Inverted flight test	20
4.5.17	Altitude accessory drive tests	20
4.5.18	Shear section strength	20
4.5.19	Mockup	20
4.5.20	Engine cranking tests	20
4.5.21	Teardown inspection	21
4.5.22	Rejection and retest	21
5.	PREPARATION FOR DELIVERY	21 - 22
5.1	Application	21
5.2	Preservation	21
5.3	Packaging	21
5.3.1	Level A	21
5.3.2	Level C	21
5.4	Packing	21
5.4.1	Level A	21
5.4.2	Level B	22
5.4.3	Level C	22
5.5	Marking	22
6.	NOTES	22 - 23
6.1	Intended use	22
6.2	Ordering data	22
6.3	Qualification	22

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Check the source to verify that this is the current version before use.

NUMERICAL INDEX (Continued)

		Page
Figure 1.	Starter no-load speed	24
Figure Z.	Pressure ratio limit to prevent starter overspeed	25
Figure 3.	Starter performance under rated inlet conditions	26
Figure 4.	Starter performance under range of inlet conditions	27
Figure 5.	Starter performance under range of inlet conditions	28
	Numerical Index	29 - 32
	Military Specification Sheet for Model ATS/AD-1	1 - 8
	Military Specification Sheet for Model ATS/AD-2	1 - 8
	Military Specification Sheet for Model ATS/AD-3	1 - 8

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